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Short prescribed exercises can quantify upper limb functioning in neurodegenerative disease

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Abstract

Background Digital health technologies (DHTs) can quantify movements in daily routines but rely heavily on participant adherence over prolonged wear times.

Methods We analyzed accelerometry data from wrist-worn devices during short at-home episodes of prescribed exercises performed by 329 individuals living with amyotrophic lateral sclerosis (ALS) in a longitudinal study. We developed an automated and interpretable signal processing method to estimate four metrics describing exercise repetitions, i.e., their count, duration, intensity, and similarity. We examined their associations with time elapsed from enrollment and ALS Functional Rating Scale-Revised (ALSFRS-R) using linear mixed effect models. We also compared them with previously validated free-living metrics that require substantial sensor wear-time. Finally, we studied how many repetitions are sufficient to determine participants' upper limb functioning.

Results Three out of four exercise metrics (all but count) demonstrated significant association with ALSFRS-R outcomes. The duration of exercise repetitions increased, while intensity and similarity of movement decreased over time (all p -value < 0.001), indicating longer but less vigorous and less consistent upper limb movements over time. Exercise intensity was determined as the most robust exercise-based predictor of changes in upper limb function, and it was comparable to free-living metrics, which required at 21 h of sensor wear time (R-squared 0.899 vs. 0.860, respectively). Sensitivity analysis indicated that as few as five exercise repetitions were sufficient to yield statistically significant associations with ALSFRS-R.

Conclusions These results suggest that prescribed exercise can effectively quantify upper limb function and track longitudinal decline comparably to free-living observation. The proposed method may serve as an alternative that decreases participation burden, increases study adherence, and extends diagnostic accessibility.

Keywords Amyotrophic lateral sclerosis, Motor function, Upper extremity, Digital health, Wrist-worn, Wearable accelerometer, Algorithm

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Background

Digital health technologies (DHTs), including smartphones and wearable devices, offer a transformative platform for frequent, remote health data collection. Recognized by the U.S. Department of Health and Human Services Food and Drug Administration (FDA) [1], DHTs hold promise for improving trial efficiency and clinical decision-making by decentralizing data collection and reducing participant burden through at-home data capture.

DHTs can collect either passive data, from devices in free-living environments, or active data, from structured tasks such as questionnaires or specific activities. Both data types can contribute to clinical outcome assessments (COAs), which measure patient experiences, functional ability, or survival, and may serve as clinically meaningful endpoints in trials. DHTs have been successfully used across diverse populations and clinical settings [2, 3], to quantify physical activity and functional status at various stages of disease severity [4–13]. Additionally, early findings suggest that DHTs may also help aid in early detection of neurological diseases, potentially opening the door for neuroprotective interventions [14].

Passive data collection with DHTs enables continuous, large-scale, and real-world monitoring with minimal participant burden [15, 16]. However, participant adherence tends to decline over time [17, 18], and identifying patterns or accounting for non-wear periods can be challenging. On the other hand, in-clinic measurements rely on brief, intermittent tests to capture changes in function. While these tests have sound psychometric properties, they require periodic clinic visits. An intermediate approach could be to use DHTs to collect data during short, standardized movement regimens at home, thus minimizing participant burden and ensuring consistent, analyzable movements.

Amyotrophic lateral sclerosis (ALS) is a neurodegenerative disease that causes progressive muscle weakness, functional decline, impaired mobility, and difficulty with daily activities [19–21]. Historically, ALS research has relied on outcome measures including quantitative strength testing, vital capacity, and the ALS Functional Rating Scale-Revised (ALSFERS-R), a questionnaire administered by trained personnel. More recently, digital COAs have been used to quantify behavioral and functional changes over time in people with ALS [10]. Passive DHT data collection has typically focused on total activity volume, real-world gait assessment, and upper limb movements [12, 22–28], often requiring prolonged wear periods for continuous data acquisition.

In this study, we introduce a new approach to quantify upper limb function in ALS based on short, prescribed exercises performed at home with wrist-worn accelerometer. From these structured tasks, we derive a set of

metrics (count, duration, intensity, and similarity) aimed to capture both the quantity and quality of repeated movements. In our main evaluation, we determine the association between these metrics, time, and ALSFERS-R. To place our work in a broader context, we also compare these metrics with a set of previously developed and validated metrics (C_{fe45} , C_{sp45} , D_{fe45} , and D_{sp45}) aimed to capture upper limb functioning in free-living setting [24, 28]. Finally, we test whether even shorter exercise tasks are useful to quantify ALS disease progression.

Methods

Population and devices.

The data used in this analysis were collected by ALS Therapy Development Institute as part of the ALS Research Collaborative Study, which up to date has enrolled over 600 individuals with ALS [29]. The program has assembled a comprehensive dataset that includes self-reported ALSFERS-R (ALSFERS-RSE) scores, digital physiological data, speech recordings, accelerometer measurements, and a biorepository of skin biopsies, whole genome sequences, and plasma samples.

We analyzed data from 329 participants with ALS enrolled between September 2014 and January 2023, wearing ActiGraph GT3X + accelerometers (ActiGraph, Pensacola, Florida) on both their dominant and non-dominant wrists. These devices recorded continuous triaxial accelerometer measurements at a sampling frequency of 30 Hz with a dynamic range of ± 6 g. In this decentralized study, devices were mailed to the participants, and they were instructed to wear the devices as much as possible for 7 days each month. During this time, participants were asked to perform three exercise sessions on alternating days consisting of five types of exercises each and annotate time at the start and finish of the entire exercise session. The upper limb exercises involved shoulder abduction and adduction, raising the limb as high and as frequently as possible within a 45-second period, followed by a 45-second rest. A YouTube video [30] was provided to guide participants through the session, which lasted approximately five minutes in total, including rest periods. Our analysis focused on the upper limb swings performed during these exercises.

Exercise quantification

Estimation of fundamental frequency

The prescribed exercises can be described in terms of their total count, average duration, intensity, and similarity between repeated movements. We quantified these metrics from the time series wearable device data, segmented into individual limb swings.

As pointed earlier, the exercise timing was self-reported by study participants with one-second accuracy. To segment limb swings, we first transformed the raw

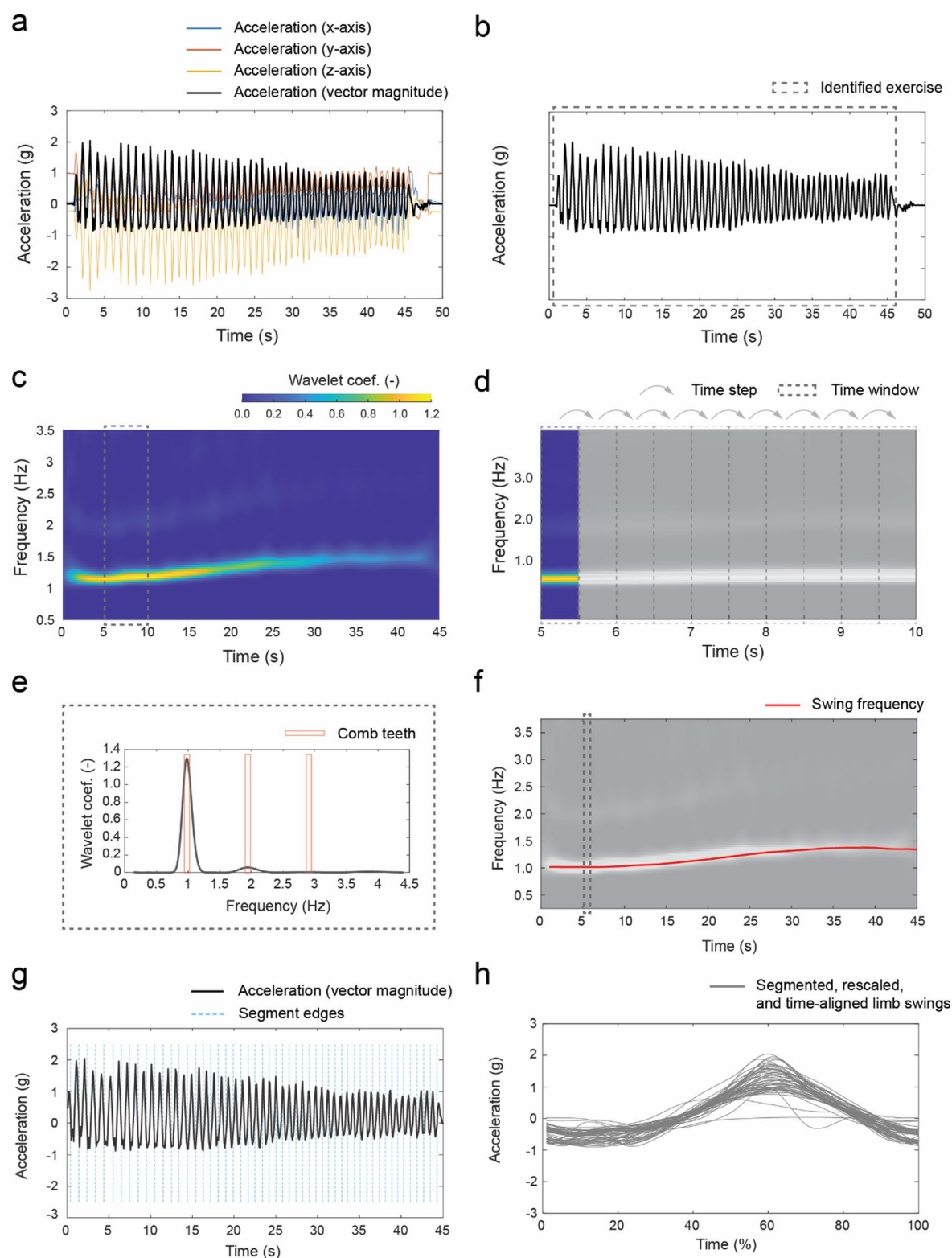


Fig. 1 (See legend on next page.)

(See figure on previous page.)

Fig. 1 Workflow for quantifying limb swings from wearable device data. **a** raw tri-axial accelerometer data were transformed into vector magnitude. **b** limb swing periods were identified using 1-second windows with normalized signal amplitude above 0.5 SD. **c** continuous wavelet transform decomposed the signal into time-frequency components to capture fundamental swing frequency and harmonics. **d** signal was split into non-overlapping windows. **e** comb function screened all possible combinations of swing frequency and its higher harmonics. **f** swing frequency was identified for each observation of the collected signal. **g** swing frequency allowed creation of a sequence of segment edges. **h** segmented signal fragments were rescaled and time aligned

tri-axial accelerometer data into univariate vector magnitude, which is robust to signal deviations and translations resulting from the non-stationary spatial orientation of the sensing device (Fig. 1a). Based on self-annotated exercise intervals from participants, limb swing periods were identified using 1-second non-overlapping windows where the maximum normalized (z-score) signal amplitude exceeded 0.5 SD for at least 10 consecutive seconds (Fig. 1b).

Next, we decomposed the signal into its time-frequency representation using continuous wavelet transform (CWT). This allowed us to uncover the dynamic characteristics of movement repetitions, with greater wavelet coefficients present at the fundamental frequency representing the limb swing frequency accompanied by its higher harmonics (i.e., peaks at multiples of the fundamental frequency) resulting from rapid changes in movement direction (Fig. 1c). The time-frequency decomposition was then split into 0.5 s non-overlapping windows (Fig. 1d), and the fundamental swing frequency was estimated from each window using a comb function (Fig. 1e) as described previously for identifying stride frequency in gait and car vibrations while driving [31, 32].

Limb swing segmentation

We used the estimated fundamental frequency (Fig. 1f) to segment limb swings. We performed this task by creating a sequence of observations corresponding to one full phase, such that each segment was bounded by adjacent observations and corresponded to an individual swing (Fig. 1g). To align these segments and estimate their similarity, we first shrunk or stretched each segmented signal (we added a margin of approximately half of segment before and after the segment of interest) to match a length of arbitrary 100 observations and shifted it by a lag corresponding to the maximum cross-correlation between the interpolated segment and a sine wave (Fig. 1h). To further improve this procedure, we identified the swing template with the highest mean cross-correlation with the remaining swing templates and repeated the alignment procedure using the identified template.

Exercise metrics

We calculated the total number of limb swings as the sum of all identified limb swing frequencies divided by two (correction for 0.5 s windows used to split the time-frequency decomposition) [33] and the swing duration

as the mean of the inverse of limb swing frequencies. Furthermore, we used segmented swing time series to calculate swing intensity and similarity. Specifically, we calculated the intensity as a mean acceleration of segmented and scaled limb swing time series, and we calculated the similarity as a mean pairwise correlation between all segmented, rescaled, and time-shifted limb swing time-series.

Free-living metrics

For comparison, we used a previously developed and validated approach for quantifying upper limb movement in free-living setting [24]. In this approach, we focused on four metrics, C_{fe45} , C_{sp45} , D_{fe45} , D_{sp45} , which estimate the total number of forearm flexions and extensions by at least 45 degrees, the total number of forearm supinations and pronations by at least 45 degrees, the average duration of the ten fastest forearm flexions and extensions by at least 45 degrees, and the average duration of the ten fastest forearm supinations and pronations by at least 45 degrees, respectively.

Statistical analysis

The main analytic dataset consisted of exercise metrics and survey scores from participants that completed at least two exercise sessions on each limb and completed at least two corresponding ALSFRS-RSE surveys. The statistics on survey submission were computed for each participant, and we described them using mean, standard deviation (SD), median, and range.

The exercise metrics were calculated for sessions with simultaneous wear of sensors on both wrists and were associated with the closest survey completed no more than 14 days from the exercise. If two surveys were equidistant from the session day, we used the earlier one.

In CWT, we used generalized Morse wavelet as the mother wavelet with symmetrical spread of coefficients in time and frequency domains [34]. The fundamental limb swing frequency was detected using a comb function with three teeth with a width of 0.1 Hz.

Pearson correlation was calculated using baseline observations, i.e., the first set of exercise metrics and ALSFRS-RSE surveys. For surveys, we considered five scores: the sum of all responses to questions 1 through 12, denoted “Q1–12”, and a sum of responses for four areas of functioning: bulbar (“Q1–3”), fine motor (“Q4–6”), gross motor (“Q7–9”), and respiratory (“Q10–12”).

The total ALSFRS-RSE is scored 0–48 with higher scores indicating higher level of function. Each of the four sub-domains is scored 0–12, with higher values indicating higher level of function in that domain [35]. To determine the consistency of each metric during the first week of data, we calculated the intraclass correlation coefficient, ICC [1, 3, 36].

Further modelling considered measures and survey responses averaged on a weekly level. We used linear mixed-effect models (LMMs) with participant-specific slope and intercept to estimate baseline and weekly change of limb swing measures and ALSFRS-R scores. Additional LMMs were used to assess the association between measures and survey responses.

For comparison, an alternative analytic dataset was derived from the same source dataset as the main analytic analysis and consisted of exercise and free-living metrics computed from days with a minimum of 21 h of sensor wear time. Sensor wear time was assessed using an established method proposed by Choi and colleagues [37].

Finally, a sensitivity analysis was performed to determine whether measures from a prespecified number of limb swings were useful to characterize upper limb disease progression. In this scenario, metrics were quantified from 5, 10, 15, and 20 initial limb swing repetitions.

Table 1 Summary on population characteristics, collected surveys, and exercise statistics

Population characteristics	
Total number of participants, N	329
Demographics	
Age, mean (SD)	54.20 (11.01)
Males, N (%)	212 (64.44)
Race, N (%)	
White	314 (95.44)
Hispanic	5 (1.52)
Asian	5 (1.52)
Black	2 (0.61)
Other	3 (0.91)
Handedness, N (%)	
Right	282 (85.71)
Left	46 (13.98)
Either	1 (0.30)
ALSFRS-RSE, mean (SD)	
Number of surveys, count	31.81 (26.64)
Time between surveys, days	37.13 (20.06)
Study participation	
Time between first and last exercise, weeks	144.60 (115.13)
Exercise, mean (SD)	
Weeks with exercise per participant, count	16.71 (16.15)
Time between weeks with exercises, weeks	6.27 (2.84)
Annotated exercise per participant, count	47.01 (47.09)
Exercise duration, non-dominant wrist, seconds	46.49 (2.29)
Exercise duration, dominant wrist, seconds	46.03 (2.11)

Metrics were then fitted using additional LMMs, as described above.

For all analysis, the statistical significance level was set at 0.05. For the coefficient of determination, R-squared, we used grouped bootstraps with 1000 iterations to estimate its 95% confidence intervals.

Results

Population characteristics

Our analysis included 329 participants with ALS who contributed a total of 15,467 epochs of accelerometer exercise data (about 200 h) during at least two self-annotated exercise sessions (Table 1). The participants were predominantly right-handed ($n=282$, 85.71%) White ($n=314$, 95.44%) males ($n=212$, 64.44%). The average age of participants at enrollment was 54.20 (SD = 11.01). On average, participants filled 31.81 (SD = 26.64) surveys every 37.13 (20.06) days and completed 47.01 (47.09) exercises over 15.40 (15.65) weeks. The average time elapsed between the first and last analyzed exercise was 144.60 (115.13) weeks. Study enrollment was presented in Fig. 1, Supplementary Materials. Exercise duration was similar for non-dominant and dominant limbs, i.e., 46.49 (2.29) and 46.03 (2.11) s, respectively. The mean baseline total ALSFRS-RSE score was 40.32 (5.31), while mean baseline Q4–6 score (representing fine motor function) was 9.50 (2.34).

Baseline associations

During the first week of observation, 259 (78.72%) participants completed three non-dominant limb exercises, 46 (13.98%) completed two, and 10 (3.04%) completed one. Fourteen participants (4.26%) did not perform any exercise sessions. The intraclass correlation coefficients, ICC [1, 3], between metrics derived from participants who completed three exercises indicated good to excellent consistency, with values of 0.819 for total limb swing count, 0.799 for duration, 0.955 for intensity, and 0.890 for similarity.

For the dominant wrist, the ICC [1, 3] values were higher for count (0.861), duration (0.840), and intensity (0.946), but not for similarity (0.808). We observed a similar distribution of exercise completion: 257 participants (78.12%) completed three dominant limb exercises, 48 (14.59%) completed two, 10 (3.04%) completed one, and 14 (4.26%) performed none.

Correlation between three of the four exercise metrics (count, duration, similarity) of the non-dominant limb and ALSFRS-RSE responses was very weak to weak, while the correlation between intensity and total score (Q1–12) and the fine motor functioning sub-domain (Q4–6) was moderate, with values of 0.43 and 0.45, respectively (Fig. 2), suggesting substantial

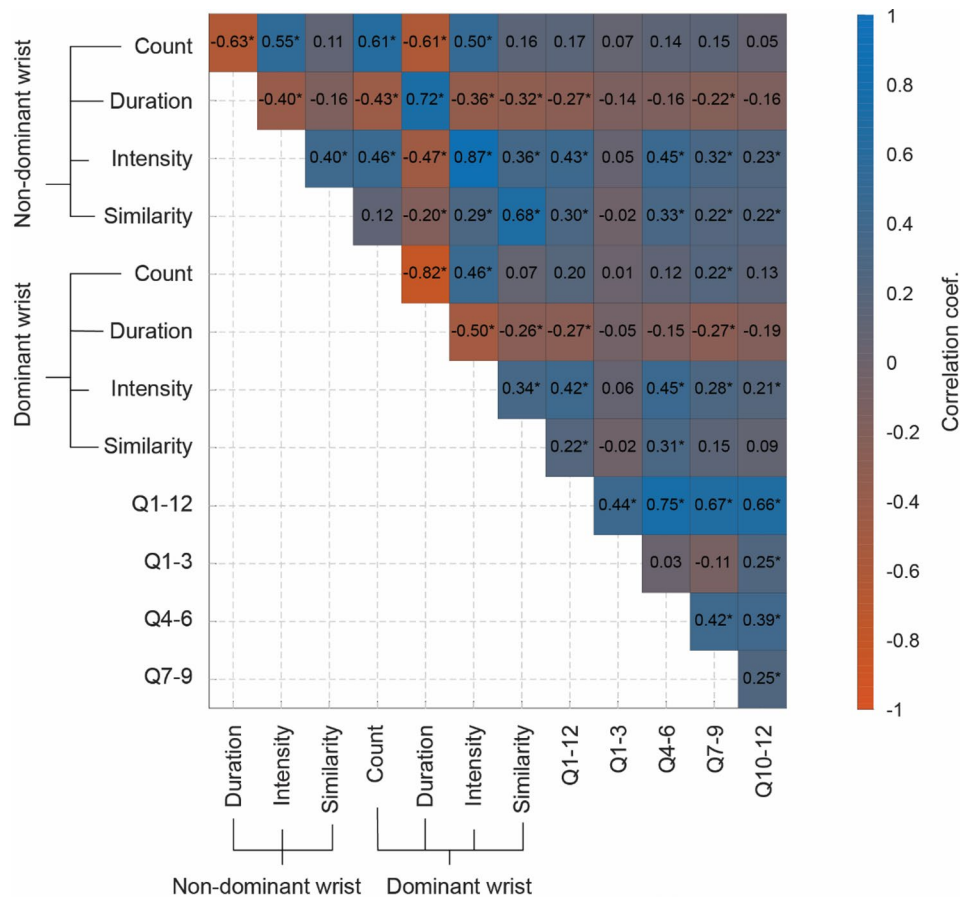


Fig. 2 Pearson correlation coefficients. Correlation coefficients were estimated between baseline exercise measures and ALSFRS-RSE total score and subdomain scores (Q1—3 bulbar, Q4—6 fine motor, Q7—9 gross motor, Q10—12 respiratory). ** indicates statistically significant correlation

between-patient heterogeneity. Similar correlation coefficients were computed for the dominant upper limb swings.

Notable correlations were identified between measures computed from each upper limb. Moderate to strong and negative correlations were identified between count and duration metrics (-0.63 and -0.82 for non-dominant and dominant upper limb, respectively), while moderate negative correlations were observed between intensity and duration metrics (-0.40 and -0.50 for non-dominant and dominant limb, respectively). Moderate positive correlations were observed between count and intensity (0.55 and 0.46 for non-dominant and dominant limb, respectively). These findings indicate that participants who performed more limb swings also executed them more quickly and vigorously, as expected. Strong to very strong positive correlations were observed between metrics from the two limbs (0.61 count, 0.72 duration, 0.87 intensity, 0.68 similarity), indicating similar within-participant upper limb functioning.

Association with time

Using LMMs, we determined that, as expected in this experimental setup, the count of limb swings remained relatively stable over time in both arms (p -value > 0.312) (Table 2). In contrast, quality metrics – duration, intensity, and similarity – demonstrated statistically significant change over time (all p -value < 0.001). Specifically, exercise duration increased, indicating slower movements, while intensity and similarity measures decreased, highlighting a decline in both vigor and consistency of limb swing repetitions. For both limbs, intensity showed the highest R-squared value, indicating that it was the most accurate and reliable measure for explaining changes in limb function over time.

A longitudinal evaluation of exercise metrics from dominant limb in one study participant over a 3-year period, using raw accelerometer data collected about every 4 weeks, demonstrated a decline in swing intensity and an increase in swing duration over time (Fig. 3). Of note, this participant also reduced the number of repetitions performed over time, deviating from the group-level trend where swing counts remained relatively stable (upper left panel in Fig. 3c). Representative patterns of

Table 2 Average baseline and weekly change in ALSFRS-RSE scores and the limb swing movements metrics

Device location	Outcome	Baseline estimate [95% CI]	Weekly change estimate [95% CI]	p-value	R-squared [95% CI]
Non-dominant wrist	Count	34.66 [33.73, 35.59]	0.002 [−0.014, 0.019]	0.773	0.758 [0.714, 0.808]
	Duration	1.408 [1.369, 1.447]	0.002 [0.001, 0.003]	5.317e-05	0.650 [0.594, 0.702]
	Intensity	0.6316 [0.5656, 0.6975]	−0.002 [−0.003, −0.001]	4.515e-06	0.882 [0.856, 0.911]
	Similarity	0.8967 [0.8831, 0.9103]	−0.001 [−0.001, −0.001]	3.931e-11	0.645 [0.607, 0.686]
Dominant wrist	Count	34.81 [33.85, 35.78]	0.008 [−0.008, 0.025]	0.312	0.771 [0.694, 0.820]
	Duration	1.408 [1.368, 1.448]	0.001 [0.000, 0.002]	0.003	0.634 [0.575, 0.718]
	Intensity	0.674 [0.6038, 0.7443]	−0.002 [−0.003, −0.001]	1.424e-06	0.873 [0.842, 0.905]
	Similarity	0.9076 [0.8957, 0.9195]	−0.001 [−0.001, −0.001]	3.291e-12	0.665 [0.612, 0.726]
ALSFRS-RSE	Q1—3	10.57 [10.34, 10.8]	−0.025 [−0.029, −0.020]	1.000e-25	0.965 [0.953, 0.972]
	Q4—6	9.61 [9.345, 9.876]	−0.039 [−0.045, −0.034]	7.497e-47	0.950 [0.942, 0.957]
	Q7—9	9.208 [8.932, 9.484]	−0.046 [−0.051, −0.041]	3.445e-61	0.955 [0.947, 0.963]
	Q10—12	11.28 [11.11, 11.45]	−0.016 [−0.019, −0.012]	8.457e-21	0.927 [0.907, 0.944]
	Q1—12	40.69 [40.09, 41.29]	−0.128 [−0.143, −0.113]	7.318e-63	0.967 [0.958, 0.975]

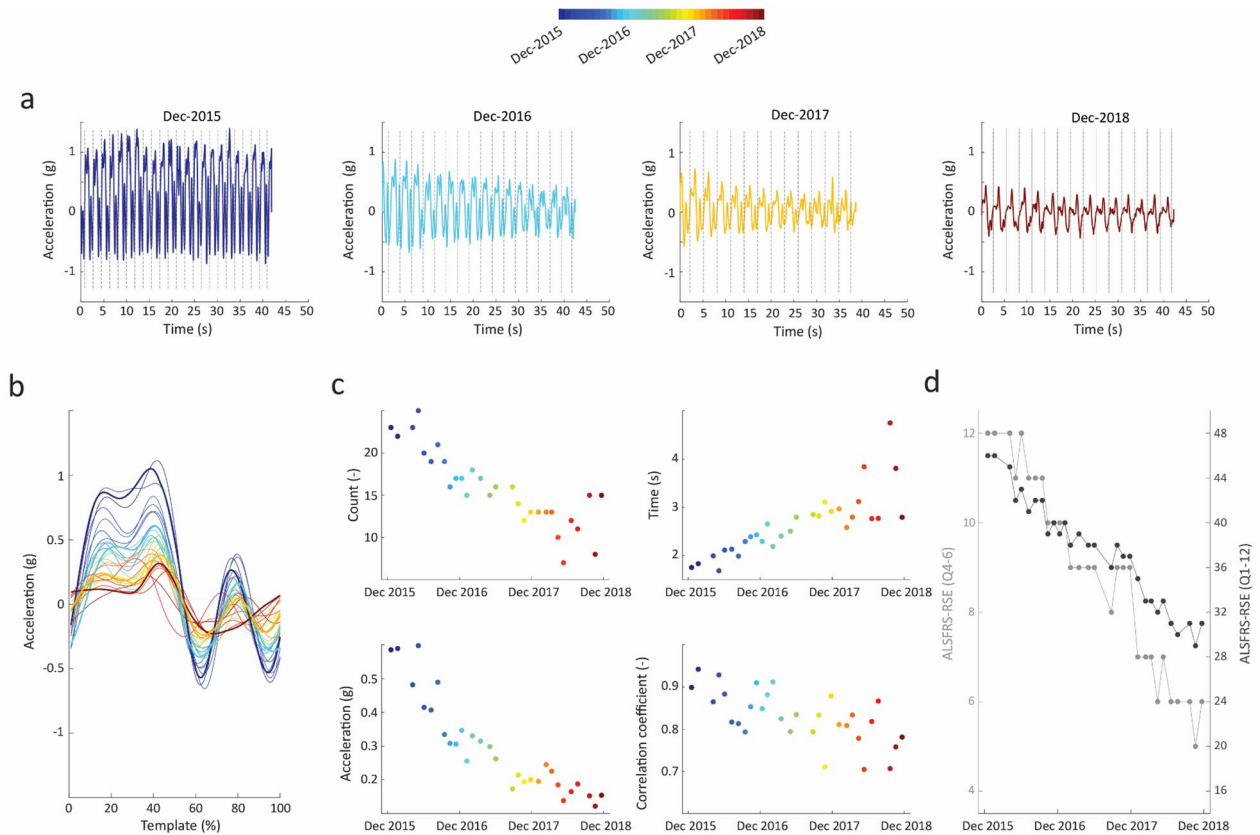


Fig. 3 Evaluation of one participant's dominant upper limb swings over 3 years. **a** on average, raw accelerometry data (vector magnitude) were collected every 6 weeks. **b** data were segmented to obtain individual limb swings templates. Templates were averaged at each time point (all limb swings during exercise, typically 30 or more) to alleviate longitudinal monitoring of disease progression. **c** we used segmented limb swings to quantify their count, duration, intensity, and similarity. **d** exercise metrics were compared with ALSFRS-RSE Q1—12 and Q4—6 responses. Data were derived from the first exercise recorded in December 2015, December 2016, December 2017, and December 2018, i.e., 37 months apart between first and last measurement. During this period, participant decreased their swing count (upper left corner of **c**), intensity (lower left corner of **c**), and similarity (lower right corner of **c**), while its duration increased (upper right corner of **c**) indicating progressing functional decline in upper limb

dominant limb exercise data are presented in the Supplementary Materials (Figures S1—S10).

Association with ALSFRS-RSE

All exercise metrics showed significant associations with fine motor function (Table 3), the total ALSFRS-RSE score (Table S1, Supplementary Materials), and the other ALSFRS-RSE subdomains (Tables S2-4, Supplementary Materials). Interestingly, similar results were computed for data collected from both wrists despite survey items such as handwriting (Q4) being specifically tied to dominant-hand activities. Among all, the greatest R-squared was noted from the intensity metrics (0.897 and 0.899 for non-dominant and dominant limb, respectively).

Comparison with free-living metrics

The comparative analysis considered data from 193 individuals who recorded at least 21 h of wear time per day during the free-living monitoring period, representing 41.5% reduction compared to the sample size in the main analytic dataset and emphasizing challenges associated with adherence related to near-continuous wear. In this subgroup, the baseline total ALSFRS-RSE score was equal to 40.03 (SD = 5.56), while Q4—6 was equal to 9.44 (SD = 2.27), similar to the original cohort.

Table 3 Model intercept and estimated average change associated with a one-point increase in ALSFRS-RSE Q4—6 using limb swing-derived outcomes of accelerometry data for PEs

Device location	Movement biomarker	Intercept [95% CI]	Slope [95% CI]	p-value	R-squared [95% CI]
Non-dominant wrist	Count	7.92 [7.42, 8.42]	0.020 [0.007, 0.032]	2.00e-03	0.814 [0.769, 0.849]
	Duration	9.41 [8.92, 9.89]	-0.585 [-0.879, -0.291]	9.74e-05	0.803 [0.766, 0.850]
	Intensity	7.61 [7.28, 7.94]	3.73 [3.08, 4.38]	2.47e-29	0.875 [0.848, 0.899]
	Similarity	5.82 [4.91, 6.73]	3.08 [2.14, 4.01]	9.69e-11	0.812 [0.745, 0.845]
Dominant wrist	Count	7.91 [7.37, 8.44]	0.019 [0.006, 0.033]	4.31e-03	0.816 [0.772, 0.861]
	Duration	9.27 [8.77, 9.77]	-0.534 [-0.847, -0.220]	8.59e-04	0.811 [0.769, 0.863]
	Intensity	7.54 [7.21, 7.87]	3.76 [3.10, 4.42]	6.93e-29	0.879 [0.860, 0.900]
	Similarity	5.62 [4.57, 6.68]	3.27 [2.19, 4.35]	3.08e-09	0.812 [0.768, 0.859]

Consistent with previous studies, free-living metrics were significantly associated with responses to the ALSFRS-RSE Q4—6 (Table 4) and Q1—12 scores (Table S5 in Supplementary Materials), regardless of device location on non-dominant or dominant wrist. Interestingly, three out of four (i.e., all except C_{fe45}) p -values estimated from free-living monitoring were greater than for the intensity measure estimated from 45-second exercises. Furthermore, the intensity metrics demonstrated higher R-squared values, indicating a stronger and more reliable fit to the observed outcomes compared to free-living metrics.

Sensitivity analysis

Metrics quantified on a limited number of limb swing repetitions demonstrated statistically significant association with the fine motor subdomain (Tables S6-S9 in Supplementary Materials) and the total ALSFRS-RSE score (Tables S10-S13 in Supplementary Materials), even using as few swings as five and regardless of device location on non-dominant or dominant wrist. Similar to the previous results, the most profound association was determined for the intensity metrics which exhibited in low p -value and both high and narrow coefficient of determination (R-squared).

The three investigated metrics were also significantly associated with time (Tables S14-17 in Supplementary Materials). Unlike the duration and similarity metrics, the intensity metric model showed a very good fit (all R -squared > 0.858) which emphasizes its robustness in estimation of disease status between participants and capacity to track disease status within participants.

Discussion

In this study, we demonstrated the feasibility of applying short, standardized, at-home assessments with DHTs to monitor upper limb function in individuals with ALS. Using data collected using wrist-worn accelerometers, we quantified four interpretable exercise metrics, i.e., count, duration, intensity, and similarity of limb swings, and compared their performance with previously validated free-living metrics, i.e., C_{fe45} , C_{sp45} , D_{fe45} , D_{sp45} . Consistent with previous studies showcasing the value of DHTs, including wearable accelerometers and smartphones [12, 24, 25, 38, 39], our findings not only highlight the effectiveness of these technologies, but also present a viable approach to minimize participant burden related to extended and continuous device wear.

To quantify exercise sessions, we developed an automated and interpretable signal processing method based on the time-frequency decomposition via continuous wavelet transformation, limb swing segmentation, and template matching. This approach enabled extraction of both quantitative (count) and qualitative (duration,

Table 4 Model intercept and estimated average change associated with a one-point increase in ALSFRS-RSE Q4—6 using limb swing-derived outcomes of accelerometry data for the free-living upper limb movements. Models were estimated using data from days with simultaneous exercise and substantial sensor wear time (≥ 21 h)

Device location	Movement biomarker	Intercept [95% CI]	Slope [95% CI]	p-value	R-squared [95% CI]
Non-dominant wrist	Count	7.95 [7.29, 8.61]	0.024 [0.008, 0.041]	4.27e-03	0.796 [0.742, 0.854]
	Duration	9.79 [9.07, 10.52]	-0.719 [-1.174, -0.265]	1.94e-03	0.806 [0.731, 0.870]
	Intensity	6.68 [6.13, 7.23]	3.75 [3.05, 4.45]	3.20e-25	0.899 [0.859, 0.923]
	Similarity	6.37 [4.92, 7.82]	2.68 [1.14, 4.23]	6.12e-04	0.798 [0.720, 0.869]
	C _{fe45}	7.03 [6.53, 7.53]	0.002 [0.001, 0.002]	4.17e-26	0.860 [0.815, 0.907]
	C _{sp45}	7.11 [6.60, 7.63]	0.001 [0.001, 0.001]	1.22e-21	0.851 [0.800, 0.892]
	D _{fe45}	10.89 [10.39, 11.40]	-3.82 [-4.62, -3.02]	1.45e-20	0.825 [0.763, 0.886]
	D _{sp45}	11.08 [10.53, 11.64]	-6.12 [-7.51, -4.73]	9.77e-18	0.830 [0.765, 0.892]
Dominant wrist	Count	8.30 [7.61, 9.00]	0.010 [-0.007, 0.027]	0.228	0.806 [0.739, 0.861]
	Duration	9.34 [8.70, 9.98]	-0.469 [-0.852, -0.087]	0.016	0.795 [0.721, 0.857]
	Intensity	6.72 [6.18, 7.25]	3.49 [2.86, 4.13]	7.31e-27	0.897 [0.865, 0.921]
	Similarity	5.68 [4.32, 7.05]	3.38 [1.96, 4.81]	3.40e-06	0.801 [0.736, 0.870]
	C _{fe45}	6.98 [6.48, 7.49]	0.002 [0.001, 0.002]	1.00e-24	0.870 [0.825, 0.909]
	C _{sp45}	7.13 [6.62, 7.64]	0.001 [0.000, 0.001]	8.84e-18	0.855 [0.808, 0.903]
	D _{fe45}	10.58 [10.14, 11.03]	-3.35 [-4.10, -2.61]	1.29e-18	0.864 [0.834, 0.901]
	D _{sp45}	10.85 [10.29, 11.41]	-5.63 [-6.93, -4.32]	4.01e-17	0.848 [0.801, 0.898]

intensity, similarity) metrics across multiple exercise sessions. Our statistical analysis demonstrated that three out of four exercise metrics, i.e., duration, intensity, and similarity, are significantly associated with longitudinal ALS disease progression, as captured by the ALSFRS-RSE. While repetition counts remained relatively stable, likely due to participants adhering to video instructions, the quality of movements declined over time, reflecting the progressive nature of ALS.

Among these, the intensity metric showed particularly strong association with both longitudinal change and survey responses, achieving higher or comparable coefficient of determination (R-squared) values relative to free-living metrics. Importantly, our findings also showed that the prolonged wear time required for free-living monitoring led to lower adherence and a substantially reduced valid sample size. Both results suggest the exercise intensity metric as a potential digital COA capable of capturing clinically meaningful changes in upper limb function.

Our sensitivity analysis indicates that as few as five repetitions of prescribed movement with either limb were sufficient to provide metrics significantly associated with fine motor function and total functional scores. This underscores the value of a wrist-worn accelerometers for capturing brief, yet meaningful, representations of upper limb function.

Several limitations of our study should be acknowledged. First, we lacked a direct method to verify participants' adherence to the prescribed exercise protocol in home settings. External factors such as potential inference or assistance from a caregiver may have influenced the collected measurements. We also had limited control over device placement; despite clear labeling for wrist placement (left wrist, right wrist), participants may have worn devices incorrectly, such as on opposite wrist or even the ankle. To address these limitations, future studies should incorporate real-time monitoring tools or adherence-promoting technologies, such as feedback systems or wearable compliance trackers, to ensure protocol fidelity. Enhanced user training and improved device design could further reduce errors related to improper wear and enhance the accuracy of data collection. Additionally, our analysis focused on a single upper limb movement (shoulder abduction and adduction) which does not fully capture the complexity of compound movements typically used in activities of daily living. Future research should expand to include other upper limb motions, such as elbow flexions and extensions, which are important for functional tasks like for self-feeding. Future work should also incorporate finger- and wrist-level metrics, potentially during meaningful functional tasks using higher-resolution or multimodal sensors. Finally, although our analyses used research-grade ActiGraph devices, future studies may consider

integrating commercially available wearables in bring-your-own-device schemes. Access to raw sensor data from such devices is becoming more feasible and could facilitate large-scale deployment while improving adherence, as participants may be more comfortable using familiar devices and further extend diagnostic accessibility and fairness.

While this study provides an initial validation for using a wearable device to monitor changes in limb function from prescribed exercises, establishing clinically meaningful outcomes that resonate with patients and clinicians remains essential [40]. This study was not designed to capture patient perspectives on clinically meaningful changes. Future studies should consider incorporating patient-reported outcome measures or other anchors to enhance the ability of DHT-derived metrics to quantify and predict functional changes meaningful to patients, researchers, and clinicians [41, 42]. Lastly, identifying the optimal types, durations, and frequencies of PEs that yield the most predictive value for clinical trials and clinical practice will be crucial, as will considering patient feedback on burden and clinical relevance to ensure adherence [43].

Conclusions

In conclusion, we demonstrated that short-duration, standardized exercise sessions using wrist-worn accelerometers may be effective tools for quantifying and monitoring upper limb disease progression in ALS. These methods have broad applicability to other conditions characterized by motor decline, offering a flexible and efficient tool for clinical trials and routine care. Combining DHTs with disease-specific quantitative assessments may enhance efficiency in clinical trials and inform clinical decision-making to support improved health outcomes, ultimately contributing to better quality of life for patients.

Abbreviations

DHT	Digital health technology
FDA	Food and Drug Administration
COA	Clinical outcome assessment
ALS	Amyotrophic lateral sclerosis
ALSFRS	R-ALS Functional Rating Scale-Revised
CIRBI	Center for IRB Intelligence
ALSFRS	RSE-Self-reported ALSFRS-R
CWT	Continuous wavelet transform
SD	Standard deviation
LMM	Linear mixed-effect model
ICC	Intraclass correlation coefficients

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12984-025-01829-z>.

Supplementary Material 1.

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Author contributions

Concept and design – MS, KMB, NC, AP, KTC, FGV, JPO, JDB. Data collection – AP, FGV. Statistical analysis – MS, JPO. Interpretation of results – MS, KMB, NC, KTC, JDB. Manuscript preparation – MS. Critical review of manuscript – KMB, NC, AP, KTC, FGV, JPO, JDB. Study supervision – JDB.

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Data availability

Data may be shared upon request and after review and approval by the owners of the data. Shared data will consider deidentified summary statistics used for analytical analysis. Related correspondence should be sent to fvieira@als.net. Data processing and statistical analyses were completed in MATLAB R2021a (MathWorks, Natick, MA). The code to estimate exercise metrics is publicly available at (https://github.com/MStrackiewicz/quantify_limb_swings). Code to estimate upper limb movement metrics used in comparative analysis has been published previously (https://github.com/MStrackiewicz/quantify_forearm_movements).

Declarations

Ethics approval and consent to participate

This study adhered to the guidelines outlined in the Declaration of Helsinki – Ethical Principles for Medical Research Involving Human Subjects. The study protocol was approved by the Institutional Review Board (ADVARA Center for IRB Intelligence (CIRBI)). All participants provided informed consent prior to any study procedures. There was no compensation for participation in the study.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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